「Commutator

「Def. Let G be a group. Given $x, y \in G$, define
$$[x,y] \overset{\text{def}}{=} x^{-1}y^{-1}xy$$
is called the Commutator of x and y. And, given two subset A and B,
$$[A,B] \overset{\text{def}}{=} \langle [a,b] \mid a \in A,\ b \in B \rangle$$
Particularly, $G' \overset{\text{def}}{=} [G,G]$ is called the derived subgroup.

「Prop. G' char G. (So $G' \trianglelefteq G$).

Proof. Given $\sigma \in \mathrm{Aut}(G)$ and $[x,y]$ for arbitrary $x,y \in G$,
$$\sigma([x,y]) = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1} \cdot \sigma(y)^{-1} \cdot \sigma(x) \cdot \sigma(y) = [\sigma(x), \sigma(y)]$$
So, $\sigma[[G,G]] = [G,G]$.

「Prop. Given $H \leq G$, $H \trianglelefteq G$ if and only if $[H,G] \leq H$

Proof. $H \trianglelefteq G \iff$ for all $g \in G$ and $h \in H$, $ghg^{-1} \in H$

$[H,G] \leq H \iff$ for all $g \in G$ and $h \in H$, $[h,g] = h^{-1}g^{-1}hg \in H$.

So proved. ∎

「Prop. Given $N \trianglelefteq G$, G/N is abelian if and only if $G' \leq N$.

Proof. G/N is abelian iff for any $xN, yN \in G/N$, $xyN = yxN$

iff for any $x,y \in G$, $x^{-1}y^{-1}xy \in N$

iff $G' \leq N$. ∎

Note: G' is the smallest normal subgroup the quotient is abelian.
In other say, G/G' is the largest abelian quotient group.

$$G' = \bigcap_{\substack{N \trianglelefteq G \\ G/N \text{ is abelian}}} N$$

「Prop. If $\varphi: G \to A$ is a Homomorphism into an abelian group A, then $G' \leq \ker \varphi$.

Proof. By the first Isomorphism Theorem, $G/\ker\varphi \cong \mathrm{Im}\,\varphi \leq A$.
Since A is abelian, so $\mathrm{Im}\,\varphi$ is also abelian. Therefore, $G' \leq \ker\varphi$. ∎

Alternatively, given $x,y \in G$, $\varphi([x,y]) = \varphi(x)^{-1}\varphi(y)^{-1}\varphi(x)\varphi(y) = 1_A$. ∎

「Homomorphism decomposition.

Let $\varphi: G \to A$ be a group Homomorphism into abelian group A.
By the first-Isomorphism theorem, $G/\mathrm{Ker}\,\varphi \cong \mathrm{Im}\,\varphi \le A$, so $G' \le \mathrm{Ker}\,\varphi$.
Therefore, $\psi: G/G' \to A;\ aG' \mapsto \varphi(a)$ is well-defined, because

$$a_1G' = a_2G' \iff a_1^{-1}a_2 \in G' \Rightarrow a_1^{-1}a_2 \in \mathrm{Ker}\,\varphi \iff \varphi(a_1^{-1}a_2) = e \iff \varphi(a_1) = \varphi(a_2)$$

Now, the onto Homomorphism, $\pi: G \to G/G';\ g \mapsto gG'$ satisfies;

$$\begin{tikzcd}
G \arrow[d, "\pi"'] \arrow[dr, "\varphi"] \arrow[d, phantom, shift left=4ex, "\circlearrowleft"] & \\
G/G' \arrow[r, "\psi"'] & A
\end{tikzcd} \qquad \varphi = \psi \circ \pi.$$

And,

$$\mathrm{Ker}\,\varphi = \varphi^{-1}[\{e\}] = \pi^{-1}[\psi^{-1}[\{e\}]] = \pi^{-1}[\mathrm{Ker}\,\psi].$$

Commutator of Typical Subgroups.

D_{2n}, the dihedral of order $2n$.

Prop. $D_{2n}' = \langle r^2 \rangle \cong \begin{cases} C_n & \text{if } n \text{ odd} \\ C_{\frac{n}{2}} & \text{if } n \text{ even} \end{cases}$

「S_n, the symmetric group on n letters

「Prop. $S_n' = A_n$.

Proof. Since $S_n/A_n \cong C_2$, abelian, so $S_n' \leq A_n$.

Debris,

$[S_3, S_3] = \langle [a,b] \mid a, b \in S_3 \rangle$

$[(1\ 2),(1\ 3)] = (1\ 2)(1\ 3)(1\ 2)(1\ 3) = (1\ 3\ 2)(1\ 3\ 2) = (1\ 2\ 3)$

$[(1\ 2),(2\ 3)] = (1\ 2)(2\ 3)(1\ 2)(2\ 3) = (2\ 3\ 1)(2\ 3\ 1) = (1\ 3\ 2)$

$[(1\ 3),(2\ 3)] = (1\ 3)(2\ 3)(1\ 3)(2\ 3) = (3\ 2\ 1)(3\ 2\ 1) = (1\ 2\ 3)$

$[(1\ 2),(1\ 2\ 3)] = (1\ 2)(1\ 2\ 3)^{-1}(1\ 2)(1\ 2\ 3) = (1\ 2)(3\ 2\ 1)(1\ 2)(1\ 2\ 3) = (1\ 3\ 2)$

$[(1\ 3),(1\ 2\ 3)] = (1\ 3)(1\ 2\ 3)^{-1}(1\ 3)(1\ 2\ 3) = (1\ 3)(3\ 2\ 1)(1\ 3)(1\ 2\ 3) = (1\ 3\ 2)$

$[(2\ 3),(1\ 2\ 3)] = (2\ 3)(1\ 2\ 3)^{-1}(2\ 3)(1\ 2\ 3) = (2\ 3)(3\ 2\ 1)(2\ 3)(1\ 2\ 3) = (1\ 3\ 2)$.

$[(1\ 2\ 3),(1\ 3\ 2)] = id$.

There fore, $[S_3, S_3] = \langle (1\ 2\ 3), (1\ 3\ 2) \rangle = \{id, (1\ 2\ 3), (1\ 3\ 2)\} = A_3$.

Alternatively, Since A_3 is normal subgroup of S_3 and $S_3/A_4 \cong C_2$, so $[S_3, S_3] \subseteq A_3$. Conversely, since $(1\ 3\ 2) \in A_3$ and $(1\ 3\ 2) = [(1\ 2),(2\ 3)] \in [S_3, S_3]$, There fore $[S_3, S_3] \supseteq A_3 = \langle (1\ 3\ 2) \rangle$.

Since $S_n = \langle (1\ 2), (1\ 2\ 3 \cdots n) \rangle$, because:

$(1\ 2 \cdots n)(1\ 2)(n \cdots 2\ 1) = (2\ 3)$

$(1\ 2 \cdots n)(2\ 3)(n \cdots 2\ 1) = (3\ 4)$

$\vdots$

$(1\ 2 \cdots n)(n-2\ \ n-1)(n \cdots 2\ 1) = (n-1\ \ n)$

and for any $1 \leq i \leq n$

$(i\ \ i+1)(i+1\ \ i+2)(i\ \ i+1) = (i\ \ i+2)$

$(i\ \ i+2)(i+2\ \ i+3)(i\ \ i+2) = (i\ \ i+3)$

$\vdots$

$(i\ \ i+j)(i+j\ \ i+j+1)(i\ \ i+j) = (i\ \ i+j)$, inductively,

$\langle (1\ 2), (1\ 2 \cdots n) \rangle$ Contains all transpositions in S_n, So it generates S_n.

Now,

$S_n' = [S_n, S_n] =$

$\ulcorner A_n$, the alternating group.

ΓQ_8. The Quaternion.